Suman Cha

Integrated M.S./Ph.D. Student in Statistics and Data Science ⋄ Yonsei University

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Website GitHub Google Scholar

EDUCATION

Yonsei University *March 2021 – February 2027 (expected)*
Integrated M.S./Ph.D. in Statistics and Data Science
Co-advisors are Prof. Hyunjoong Kim and Prof. Ilmun Kim.

Yonsei University *March 2015 – February 2021*
B.S. in Applied Statistics

RESEARCH AND INDUSTRY EXPERIENCE

KRAFTON *June 2025 – March 2026*
AI Research Intern, User Modeling AI
Conducted applied machine learning research for user modeling.

Yonsei University *January 2025 – December 2026*
Researcher, Water-Treatment Research
Conducted statistical and machine learning research for water-treatment data.

Yonsei University *September 2022 – February 2025*
Researcher, BRL Two-Sample Testing Project
Conducted research on conditional two-sample testing.

PUBLICATIONS AND MANUSCRIPTS

[1] **Suman Cha***, Seongchan Lee*, Dohyeon Ko*, and Hyunjoong Kim†, "FUSE: Feature-Wise Unified Specialization with Cross-Column Exchange for Mixed-Type Tabular Flow Matching." *Submitted to AAAI 2027.*
[Code]

[2] Jungmin Kim*, **Suman Cha***, Seunghan Lee*, Seunghyun Kim, Junekyu Park, Jiyoung Lim, Anna Lee, and Hyuck Lee†, "Real-Time Win Probability Prediction in Battle Royale Games via Survival Analysis." *Submitted to KDD 2027 Applied Data Science Track.*

[3] **Suman Cha***, Seongchan Lee*, Antonin Schrab, and Ilmun Kim, "More Permutations Do Not Always Increase Power: Non-monotonicity in Monte Carlo Permutation Tests." *Submitted to Statistical Science.*
[arXiv · Code]

[4] **Suman Cha** and Hyunjoong Kim, "Learning Majority-to-Minority Transformations with MMD and Triplet Loss for Imbalanced Classification." *Information Sciences 123929 (2026).*
[Publisher · arXiv · Code]

[5] Seongchan Lee*, **Suman Cha***, and Ilmun Kim, "General Frameworks for Conditional Two-Sample Testing." *Submitted to Biometrika.*
[arXiv · Code]

* Equal contribution. † Corresponding author.

OPEN RESEARCH SOFTWARE

Generative and Imbalanced Learning

- **FUSE**. Reproducibility pipeline with SHA-256-verified data reconstruction and command-line workflows for training, reproduction, dependence studies, replay, and evaluation. (Publication 1.)
- **MOMS**. Parametric majority-to-minority transformation using MMD for global alignment and triplet loss for local boundary awareness, evaluated on 29 datasets. (Publication 4.)

Statistical Testing

- **Cond2ST**. R package and reproducibility code for two general conditional two-sample testing approaches based on conditional independence tests and density-ratio testing. (Publication 5.)
- **mc-perm-power**. R, Rcpp, and C++ reproducibility code for the non-monotone unconditional power and integer-threshold sawtooth behavior of standard non-randomized Monte Carlo permutation tests. (Publication 3.)

SELECTED TALKS AND AWARD

Talks

2025 **Oral presentation**, JSM 2025.
General Frameworks for Conditional Two-Sample Testing

2024 **Oral presentation**, KDMS 2024.
Contrastive Learning for Two-sample Tests

Award

2024 **Best Paper Award**, KDMS 2024.

TECHNICAL SKILLS

Programming and ML	Python, PyTorch, R, C++
Data platforms	Databricks, PySpark